



1. Let $O = 2$, $M = 3$, and $C = 4$. Find $\frac{O}{M} \frac{M}{C}$.

- (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) 2 (E) 3

2. If $ab = 40$, $cd = 42$, $ac = 210$, find db .

- (A) 2 (B) 4 (C) 6 (D) 7 (E) 8

3. The numbers $\sqrt{13}$, $\sqrt[6]{13}$, form the first and third terms of a geometric sequence. Find the fourth term.

- (A) 1 (B) $\sqrt[13]{13}$ (C) $\sqrt[12]{13}$ (D) $\sqrt[10]{13}$ (E) $\sqrt[9]{13}$

4. A user of AoPS has a ratio of 2 times as many posts than upvotes. If he were to make a new post, he would need 77 more upvotes to achieve a ratio of 2 times as many upvotes than posts. Find the number of posts he currently has.

- (A) 25 (B) 33 (C) 50 (D) 66 (E) 77

5. Mom was baking cookies. The number of cookies she baked on the weekdays (Monday through Friday) is a multiple of 11. The number of cookies she baked over the weekend is a multiple of 17. The (nonzero) number of cookies she baked across the entire week is a multiple of 16. Find the smallest possible value of this quantity.

- (A) 64 (B) 80 (C) 96 (D) 112 (E) 128

6. Find the number of ordered pairs of positive integers (a, b) satisfying $1 < a < b < a + 4 < 20$.

(A) 36 (B) 42 (C) 48 (D) 49 (E) 64

7. How many 5-digit positive integers (without leading zeroes) are equal to 7000 times the sum of their digits?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

8. Jared has a bag with at least 7 fruits, each of which is an orange, banana, or apple (at least one fruit of each type). For every 7 fruits Jared selects out of the bag, there will always be at least one orange and at least 4 fruits of the same type. Find the number of possible amounts of apples in the bag.

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

9. Ana draws the vertices of a square and the midpoints of its sides on a paper, for 8 points in total. What is the minimum amount of circles Bob must draw so that for all pairs of points among the set Ana drew, there exists a circle that Bob drew that both points lie on?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

10. Define $\triangle ABC$ with $\angle A = 90^\circ$, $AB = 3$ and $AC = 4$. Define square $BCDE$ outside ABC . If the bisector of $\angle A$ intersects DE at F , find DF .

(A) $\frac{15}{8}$ (B) $\frac{15}{7}$ (C) $\frac{5}{2}$ (D) $\frac{20}{7}$ (E) 3

11. If a, b are the two roots of $x^2 - 6x + 7 = 0$, find $\arctan a + \arctan b$.

(A) $-\frac{\pi}{4}$ (B) 0 (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$ (E) $\frac{3\pi}{4}$

12. Find the sum of the digits of the smallest positive integer greater than the product of 300 and its largest prime divisor. ("its largest prime divisor" is referring to the chosen positive integer itself and its largest prime divisor, not to 300's largest prime divisor.)

(A) 7 (B) 14 (C) 15 (D) 18 (E) 21

13. Find the number of two digit positive integers n where $\lfloor \sqrt{n} \rfloor$ divides n .

- (A) 6 (B) 7 (C) 20 (D) 21 (E) 27

14. Define quadrilateral $ABCD$ where B and D are right, $AB = AD$, and $CB = CD$. The areas of the largest semicircles centered on BD that can fit inside ABD and CBD are 40π and 32π . Find BD .

- (A) 12 (B) $12\sqrt{3}$ (C) 24 (D) $16\sqrt{3}$ (E) 48

15. Let $a, b \geq 0$ be integers where $2^a 3^b < 10^4$. Find the largest possible value of $3a + 5b$.

- (A) 39 (B) 40 (C) 41 (D) 42 (E) 43

16. Let $ABCDE$ be a convex pentagon with $AB = AE = 7$, $CB = CD = 6$, $\angle A = \angle C = 90^\circ$, and $AC = 12$. Find $ABCDE$'s area.

- (A) 70 (B) 72 (C) 75 (D) 80 (E) 84

17. Find the number of sequences of integers $a_0, a_1, \dots, a_7$ between 0 and 3 inclusive where $\sum_{i=0}^7 2^i a_i = 640$

- (A) 62 (B) 63 (C) 124 (D) 125 (E) 126

18. Find the sum of digits of the 100th smallest positive integer n where the floors of the roots of $x^2 - 4x - n$ multiply to $1 - n$.

- (A) 4 (B) 5 (C) 10 (D) 11 (E) 19

19. Owen rolls a die repeatedly until he rolls 2 numbers in a row that sum to an integer greater than 10. Find the probability his last roll is a 6.

- (A) $\frac{1}{2}$ (B) $\frac{7}{12}$ (C) $\frac{5}{8}$ (D) $\frac{2}{3}$ (E) $\frac{11}{16}$

20. Define equilateral triangle $\triangle ABC$ with point P inside. Let E, F be the feet of P to CA, AB respectively. If $PE = 1$, $PF = 2$ and $\angle APB = 120^\circ$, find the area of $\triangle ABC$.

$$(A) \frac{46}{\sqrt{3}} \quad (B) 27 \quad (C) 16\sqrt{3} \quad (D) 28 \quad (E) \frac{49}{\sqrt{3}}$$

21. A hunter and an invisible rabbit play a game on the number line. First, the rabbit materializes on an integer between 1 and 100 inclusive which is unknown to the hunter. Each day thereafter, the hunter will guess the position of the rabbit. If he is wrong, the rabbit then hops to the next integer to the right; unless it is at 100, then it stays in place. What is the least number of days the hunter needs to guarantee to be able to catch the rabbit?

$$(A) 49 \quad (B) 50 \quad (C) 51 \quad (D) 99 \quad (E) 100$$

22. What is the sum of the digits of the largest positive integer N where the equation $9a + 10b + 11c = N$ has at most 1 solution over all positive integers a, b, c ?

$$(A) 11 \quad (B) 12 \quad (C) 13 \quad (D) 14 \quad (E) 15$$

23. Define tetrahedron $ABCD$ with $AB = CD = 13$, $BC = DA = 14$, and $AC = BD = 15$. Find the absolute value of the cosine of the angle the faces ABC and BCD make with each other.

$$(A) \frac{11}{36} \quad (B) \frac{23}{72} \quad (C) \frac{1}{3} \quad (D) \frac{13}{36} \quad (E) \frac{3}{8}$$

24. Ana writes 10 ones in a row with 9 spaces between them, as shown.

1 O 1 O 1 O 1 O 1 O 1 O 1 O 1

In each space, Ana has a half chance of writing a plus sign or leaving it blank. Then she reads the expression as a sum of base 2 integers. For example, if pluses are placed in the 3rd and 8th spaces, the sum is $111 + 11111 + 11$, or $7 + 31 + 3 = 41$ in base 10. Find the expected value of the sum in base 10.

$$(A) 31.5 \quad (B) 32 \quad (C) 32.5 \quad (D) 55 \quad (E) 56$$

25. The graph of $y = x^3 + 7x^2 - 10$ intersects the graph of $x^2 + y^2 = 100$ at several points. Find the sum of coordinates of the centroid of the set of these points. (The centroid of a set S of points is a point with x coordinate as the average of the x coordinates over the points in S , and with y coordinate as the average of the y coordinates over the points in S .)

$$\text{(A)} -\frac{14}{5} \quad \text{(B)} -\frac{7}{3} \quad \text{(C)} -\frac{4}{3} \quad \text{(D)} -\frac{4}{5} \quad \text{(E)} -\frac{2}{3}$$

Time limit: 75 minutes